

■ NGO FOOD WASTAGE REDUCTION

From Idea to Working Product — Hackathon Edition

India · 2026 | React.js · FastAPI · PostgreSQL · Firebase

Built for Impact · Optimized for Hackathons · Ready to Scale

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1. Project Overview & Problem Statement

The NGO Food Wastage Reduction System is a technology platform that bridges the gap between food donors (restaurants, hotels, events, households) and NGOs. The platform enables real-time reporting, collection, and distribution of surplus food — while also intelligently handling expired food through composting, animal shelter routing, and biogas flagging.

India wastes approximately 68–70 million tonnes of food annually — enough to feed 200 million people. At the same time, over 190 million Indians remain undernourished. This project directly attacks that gap using technology.

Key Problems Being Solved

- Large volumes of edible surplus food go to waste daily at restaurants, weddings, and events
- NGOs have no real-time visibility into where surplus food is available
- No structured communication channel between food donors and relief organizations
- Expired/near-expiry food has no alternative routing — it simply gets discarded
- Volunteer coordination for pickups is manual and inefficient

Unique Value Proposition

Unlike existing platforms (Zomato Feeding India, No Food Waste), this system introduces an **Expired Food Handling Module** — routing non-distributable food to composting units, animal shelters, or biogas plants. This makes the platform a complete food lifecycle manager, not just a redistribution tool.

2. Market Research & Competitive Landscape

Platform	Strength	Gap / Weakness	Our Edge
Zomato Feeding India	Scale, tech infra	No expired food handling	Expired food module
No Food Waste (Coimbatore)	Strong volunteer network	Limited to South India	Scalable architecture
Robin Hood Army	Community driven	Manual coordination	Real-time notifications
FoodCloud	NGO connections	No expiry tracking	AI expiry prediction

■ Judges WILL ask about existing solutions. Know these competitors and always pivot to your Expired Food Module and real-time coordination as differentiators.

3. System Architecture

The system follows a **3-tier architecture**: a React.js frontend, a FastAPI backend, and a PostgreSQL + Firebase data layer. All components are containerized with Docker for easy deployment.

Layer	Technology	Responsibility
Presentation	React.js + Tailwind	Donor portal, NGO dashboard, admin panel, mobile-responsive UI
API Gateway	FastAPI (Python)	REST endpoints, authentication, business logic, ML model serving
Real-time	Firebase Realtime DB	Live notifications, pickup status updates, volunteer tracking
Primary DB	PostgreSQL	Users, donations, pickups, NGO records, food classification
Auth	Firebase Auth	OTP-based phone login (India-optimized), OAuth2
Storage	Cloudinary / S3	Food images uploaded by donors
Maps	Google Maps API	Donor location, NGO routing, volunteer navigation
Notifications	FCM + MSG91	Push alerts to NGOs, SMS to donors without smartphones

4. Tech Stack — Detailed Breakdown

4.1 Frontend — React.js

React.js provides a component-based, fast-rendering UI. Combined with Tailwind CSS, you get a professional interface without writing custom CSS from scratch.

- **React.js 18** — Core framework with hooks-based state management
- **Tailwind CSS** — Utility-first styling, responsive by default
- **React Router v6** — Client-side navigation between donor/NGO/admin views
- **Axios** — HTTP client for API calls
- **React Query** — Server state management and caching
- **Leaflet.js** — Open-source map rendering (free alternative to Google Maps)
- **Chart.js** — Dashboard analytics (donations rescued, kg saved, etc.)

4.2 Backend — FastAPI (Python)

FastAPI is the fastest Python web framework. It auto-generates API documentation (Swagger UI at /docs), has native async support, and makes it easy to plug in scikit-learn ML models.

- **FastAPI** — REST API framework with automatic OpenAPI docs
- **SQLAlchemy** — ORM for PostgreSQL database operations
- **Pydantic v2** — Data validation and serialization
- **Alembic** — Database migrations
- **python-jose** — JWT token authentication
- **scikit-learn** — ML model for expiry prediction
- **Celery + Redis** — Background task queue (scheduled pickup reminders)

4.3 Database — PostgreSQL

PostgreSQL handles structured relational data. Firebase handles real-time pub/sub. Never store real-time state in PostgreSQL — that's Firebase's job.

4.4 Mobile Strategy

For hackathon: Build a PWA (Progressive Web App) using React. This works on mobile browsers without requiring App Store deployment. Post-hackathon: Migrate to React Native for a native iOS/Android app — code reuse is ~70%.

5. Database Schema Design

Core tables and their key columns:

users

```
id, name, phone, email, role (donor/ngo/admin/volunteer), location_lat, location_lng, created_at
```

food_donations

```
id, donor_id, food_name, quantity_kg, food_status (fresh/near_expiry/expired), expiry_date, pickup_address, pickup_time, image_url, status (pending/accepted/collected/distributed), created_at
```

ngos

```
id, user_id, ngo_name, registration_number, service_area, capacity_kg, verified
```

pickups

```
id, donation_id, ngo_id, volunteer_id, scheduled_time, actual_pickup_time, status
```

volunteers

```
id, user_id, ngo_id, availability, total_pickups_done
```

expired_food_routing

```
id, donation_id, route_type (compost/animal_shelter/biogas), partner_id, status
```

partners

```
id, name, type (compost_unit/animal_shelter/biogas_plant), location_lat, location_lng, contact
```

notifications

```
id, user_id, type, message, read, created_at
```

analytics

```
id, date, total_donations_kg, total_distributed_kg, expired_routed_kg, ngos_active
```

6. Core Features & Modules

Donor Registration & Login

- Phone OTP login via Firebase Auth (no password needed — India-friendly)
- Profile: name, location, type (restaurant/household/event organizer)
- Donation history dashboard

Food Donation Submission

- Upload: food name, quantity, photo, status (fresh/near-expiry/expired)
- Expiry date picker with auto-flag if within 48 hours
- Location auto-detected via browser geolocation or manual pin on map
- Pickup window: donor sets available time slots

NGO Dashboard

- Real-time feed of available donations within service radius
- One-click accept request
- Assign pickup to available volunteer
- Track ongoing pickups on map

Real-Time Notifications

- FCM push notification when new donation matches NGO service area
- SMS fallback via MSG91 for NGOs without smartphones
- Donor gets confirmation when NGO accepts their request

Pickup Management

- Volunteer receives navigation link (Google Maps deeplink)
- Status updates: En Route → Arrived → Collected
- NGO marks food as distributed with quantity confirmation

Analytics Dashboard

- Total kg rescued this month
- Number of meals served
- CO2 emissions prevented (calculate: 1 kg food waste = 2.5 kg CO2)
- Top donors leaderboard

7. Expired Food Handling Module

This is your BIGGEST differentiator. No major competitor handles expired food systematically. Lead with this in your hackathon pitch.

When a donor marks food as **expired**, the system does NOT route it to human distribution. Instead, it activates the Expired Food Handling Module with three routing options:

Route	When to Use	Partners	Outcome
Composting	Organic waste, cooked food	Local farmers, municipal compost units	Organic fertilizer
Animal Shelter	Dry food, grains, slightly stale	Cattle shelters, animal NGOs, stray feeding orgs	Animal nutrition (follow FSSAI guidelines)
Biogas Plant	Any organic food waste in bulk	Municipal biogas plants, large NGO facilities	Biogas / cooking fuel / electricity

Expiry Prediction Feature (AI Layer)

Before food even expires, the system sends proactive alerts:

- Donor enters expiry date at donation time
- System sends 'Donate before expiry' alert 48 hours before expiry date
- If no pickup in 12 hours before expiry, auto-suggest expired food routing
- ML model (trained on historical data) can predict likely expiry per food category

■ **FSSAI Compliance: Never route expired food to human distribution. Always check smell, colour, and condition before animal shelter donation. Maintain hygiene standards throughout.**

8. Development Roadmap — Phase by Phase

Phase 1 — Foundation (Days 1–2)

- Set up React.js project with Tailwind CSS
- Initialize FastAPI backend with PostgreSQL connection
- Implement Firebase Auth (phone OTP login)
- Create user registration: donor and NGO roles
- Design and implement core database schema (users, food_donations, ngos)
- Build basic food donation form with location input

Phase 2 — Core Features (Days 3–5)

- Build NGO dashboard with donation feed
- Implement real-time notifications using Firebase
- Create pickup management flow (accept → assign → collect → distribute)
- Integrate Google Maps API for location visualization
- Build donor dashboard with donation history
- Add image upload via Cloudinary

Phase 3 — Differentiators (Days 6–8)

- Build Expired Food Handling Module with 3 routing options
- Implement expiry date tracking and 48-hour alert system
- Add food status classification (fresh/near-expiry/expired) in UI
- Build partner management (composting units, animal shelters, biogas plants)
- Create SMS notification via MSG91 for non-smartphone NGOs

Phase 4 — Polish & Deploy (Days 9–10)

- Build analytics dashboard with kg rescued, meals served, CO2 saved
- Add donor leaderboard
- Deploy frontend on Vercel, backend on Railway
- Run end-to-end testing
- Prepare hackathon demo with test data
- Create pitch deck and demo video

9. API Endpoints Reference

All endpoints are prefixed with **/api/v1**. FastAPI auto-generates interactive docs at **/docs** (Swagger UI) — use this during development.

Auth

Method	Endpoint	Description
POST	/auth/send-otp	Send OTP to phone number
POST	/auth/verify-otp	Verify OTP, return JWT token
POST	/auth/logout	Invalidate token

Donations

Method	Endpoint	Description
GET	/donations	List all available donations (with filters)
POST	/donations	Create new donation (donor only)
GET	/donations/{id}	Get donation details
PATCH	/donations/{id}/status	Update donation status
DELETE	/donations/{id}	Cancel donation (donor only)

Pickups

Method	Endpoint	Description
POST	/pickups	NGO accepts a donation
PATCH	/pickups/{id}/assign	Assign volunteer to pickup
PATCH	/pickups/{id}/status	Update pickup status (en-route/collected)

Expired Food

Method	Endpoint	Description
POST	/expired-food/route	Route expired food to partner (compost/animal/biogas)
GET	/expired-food/partners	List available partners by type and location

Analytics

Method	Endpoint	Description
GET	/analytics/summary	Overall stats (kg rescued, meals, CO2)

GET	/analytics/leaderboard	Top donors this month
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10. Frontend UI Guide

Page	Contents
Landing Page	Hero with impact stats (kg food rescued, people fed), CTA buttons for Donor and NGO signup, brief how-it-works section
Donor Dashboard	My active donations, donation history, quick-add donation button, impact counter (kg donated, meals enabled)
New Donation Form	Food name + photo upload, quantity slider, food status radio (Fresh/Near-Expiry/Expired), expiry date picker, location pin, pickup time slots
NGO Dashboard	Real-time donation feed sorted by proximity, filter by food type/status, map view of donations, active pickups list
Pickup Tracker	Live status: Pending -> Accepted -> En Route -> Collected -> Distributed, volunteer contact, map with route
Expired Food Handler	Triggered when food status = Expired, three routing cards (Compost/Animal/Biogas), partner selection, submit routing request
Analytics Page	Big number cards (kg rescued, meals, CO2 saved), bar chart (weekly donations), pie chart (food types), donor leaderboard table
Admin Panel	User management, NGO verification, partner approval, platform-wide analytics, flagged donations

Design tip: Use a green (#2E7D32) and white color scheme throughout. Keep the UI minimal and mobile-first. Indian NGO workers often use low-end Android devices on 4G.

11. Deployment Guide

For hackathon, prioritize getting a live URL. Judges expect to see a deployed demo, not just a localhost demo.

Service	Platform	Free Tier	Steps
Frontend	Vercel	Yes (unlimited)	Connect GitHub repo, auto-deploys on every push
Backend API	Railway	Yes (\$5 credit)	Deploy FastAPI via Dockerfile or Nixpacks
PostgreSQL	Railway / Supabase	Yes	One-click Postgres instance, get connection URL
Firebase	Firebase (Google)	Yes (Spark plan)	Enable Auth + Realtime DB from Firebase Console
Images	Cloudinary	Yes (25GB)	Use Cloudinary SDK in FastAPI for upload

Environment Variables (.env)

```
DATABASE_URL, FIREBASE_PROJECT_ID, FIREBASE_PRIVATE_KEY, CLOUDINARY_URL,  
GOOGLE_MAPS_API_KEY, MSG91_AUTH_KEY, SECRET_KEY, REDIS_URL
```

12. Hackathon Pitch Strategy

A great project with a poor pitch loses. Follow this structure:

Time	Slide/Section	What to Say
0:00 – 0:30	Hook	Start with a stat: '68 million tonnes of food wasted in India annually while 190 million go hungry. We fix that.'
0:30 – 1:30	Problem	Show the gap between donors and NGOs. Use a real scenario: a wedding hall with 100 kg of leftover food and no one to call.
1:30 – 3:00	Demo	Live demo: donor submits food -> NGO gets notification -> pickup tracked. Show the expired food module separately.
3:00 – 4:00	Tech Stack	Quickly mention React, FastAPI, PostgreSQL, Firebase. Show your architecture diagram.
4:00 – 4:30	Impact	Show analytics: kg rescued, meals served, CO2 saved. Make the numbers visible and large.
4:30 – 5:00	Differentiation	Land the expired food module hard. Mention composting, animal shelters, biogas. No competitor does this.

■ Anticipate these judge questions: (1) How is this different from Zomato Feeding India? (2) How do you ensure food safety? (3) How do you onboard NGOs? — Have crisp answers ready.

13. Future Enhancements

Mention 2–3 of these in your pitch to show you've thought about the product roadmap:

React Native Mobile App

Cross-platform iOS/Android app with 70% code reuse from React web

AI Demand Prediction

ML model predicts food availability spikes (festivals, weddings) so NGOs prepare in advance

GPS Live Tracking

Real-time volunteer location tracking for NGO coordinators

Rating & Trust System

Donors and NGOs rate each interaction; builds accountability

Government API Integration

Connect with Smart City portals and municipal compost programs

Carbon Credit Calculation

Convert CO2 saved into carbon credits for corporate CSR programs

WhatsApp Bot Integration

Donors can submit food details via WhatsApp (ultra-low friction for India)

Blockchain Donation Trail

Immutable record of food donated → collected → distributed for NGO auditing

Good luck at your Hackathon! Build with impact. Ship with pride. ■